

GF Wikidata

Large-scale grammars for natural language generation

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GF Summer School 2025

Types in GF

What is the type of:

1. `42 : ?`
2. `"alphabet" : ?`
3. `{s="monkey"; n=Sg} : ?`
4. `{s:Str, g:Gender} : ?`
5. `{n:Number, g:Gender} : ?`
6. `\x -> x : ?`
7. `\x -> x.s : ?`

Polymorphism

In Haskell:

`(\x -> x) :: a -> a` 😊

In standard GF

`oper id = \x -> x ;`

Happened in operation id

cannot infer lntype of `\x -> x`

Polymorphism in GF 4.0

```
oper id : ({a} : Type) -> a -> a  
        = \{a},x -> x ;
```

- $(\{a\} : \text{Type}) \rightarrow a \rightarrow a$ – a function with two arguments where the value of the first specifies the type of the second and the result.
- $\{\dots\}$ declares the argument as implicit, meaning that the programmer don't need to specify it
- Examples:

```
<id "test" : Str>  
<id {s="rose"; n=Sg} : {s:Str; n:Number}>  
<id {Int} : Int -> Int>
```

Polymorphism and records

In standard GF:

```
oper foo = \x -> x.s ;
```

Happened in operation foo
cannot infer litype of \x -> x.s

In GF 4.0:

```
oper id : ({a} : Type) -> {s : a} -> a  
        = \{a},x -> x.s ;
```

Why not:

```
oper id : ({a} : Type) -> {s : a; n : Number} -> a  
        = \{a},x -> x.s ;
```

Subtyping

$\{s : \text{Str}; n : \text{Number}\} < \{s : \text{Str}\}$

$\text{PType} < \text{Type}$

$\text{Ints } n < \text{Ints } m \text{ iff } n \leq m$

Type Errors in the RGL

In Different Functors and Resource Modules:

Number : Type

Agr : Type

{s : Agr => Str} – is not permitted

In Latin:

QClause = {s : C.Tense => Anteriority => C.Pol => QForm => Str} ;

Tense = {s : Str ; t : R.Tense} ;

Pol = {s : Str ; p : R.Polarity} ;

Type Error in Formal.gf

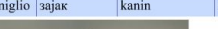
```
Prec : PType = Ints 4 ;
```

```
nextPrec : Prec -> Prec = \p -> case <p : Prec> of {  
    4 => 4 ;  
    n => Predef.plus n 1  
};
```

```
Predef.plus : Int -> Int -> Int
```

Bridging Wikidata

Linking with Wikidata

Abstract	Afrikaans	Bulgarian	Catalan	Danish	Dutch	English	French	German	Hungarian	Icelandic	Italian	Macedonian	Norwegian Nynorsk	Norwegian Bokmål	Polish	Portuguese	Romanian	Russian	Slovenian	Spanish	Swedish	Turkish	f
1. any of various burrowing animals of the family Leporidae having long ears and short tails; some domesticated and raised for pets or food																							
🐇rabbit_1_N	konyn	заяц	conill	kanin	konijn	rabbit	lapin	Kaninchen	nyúl	kanína	coniglio	zajak	kanin	kanin	królik	lebre	iepure	кролик	kunec	conejo	kanin	tavşan	
																							



European rabbit

文 92 languages

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This article is primarily concerned with the wild animal. For detailed information on domesticated varieties, see [Domestic rabbit](#). For general information on all rabbit species, see [Rabbit](#).

The **European rabbit** (*Oryctolagus cuniculus*) or **coney**^[5] is a **species** of **rabbit** native to the **Iberian Peninsula** (**Spain**, **Portugal** and **Andorra**) and southwestern **France**.^[3] It is the only living species in *Oryctolagus*, a genus of lagomorphs. The average adult European rabbit is smaller than the **European hare**, though size and weight vary with habitat and diet. Due to the European rabbit's history of **domestication**, **selective breeding**, and introduction to non-native habitats, wild and domesticated European rabbits across the world can vary widely in size, shape, and color.

European rabbits prefer **grassland** habitats and are **herbivorous**, mainly feeding on grasses and leaves, though they may supplement their diet with berries, tree bark, and field crops such as **maize**. They are prey to a variety of **predators**, including **birds of prey**, **mustelids**, **cats**, and **canids**. The European rabbit's main defense against predators is to run and hide, using vegetation and its own burrows for cover. It is well known for digging networks of **burrows**, called **warrens**, where it spends most of its time when not feeding. The European rabbit lives in **social groups** centered around territorial females. European rabbits in an established social group will rarely stray far from their warren, with female rabbits leaving the warren mainly to establish nests where they will raise their young. Unlike **hares**, rabbits are born blind and helpless, requiring maternal care until they leave the nest.

The European rabbit has had major agricultural and biological impacts as an **invasive species**, and has been hunted and raised as a food source since **medieval times**. It is the only domesticated species of rabbit, and **all known breeds of rabbit** are its descendants. It has often been introduced to exotic locations as a food source or for sport hunting. Starting from the first century BCE, it has been introduced to at least

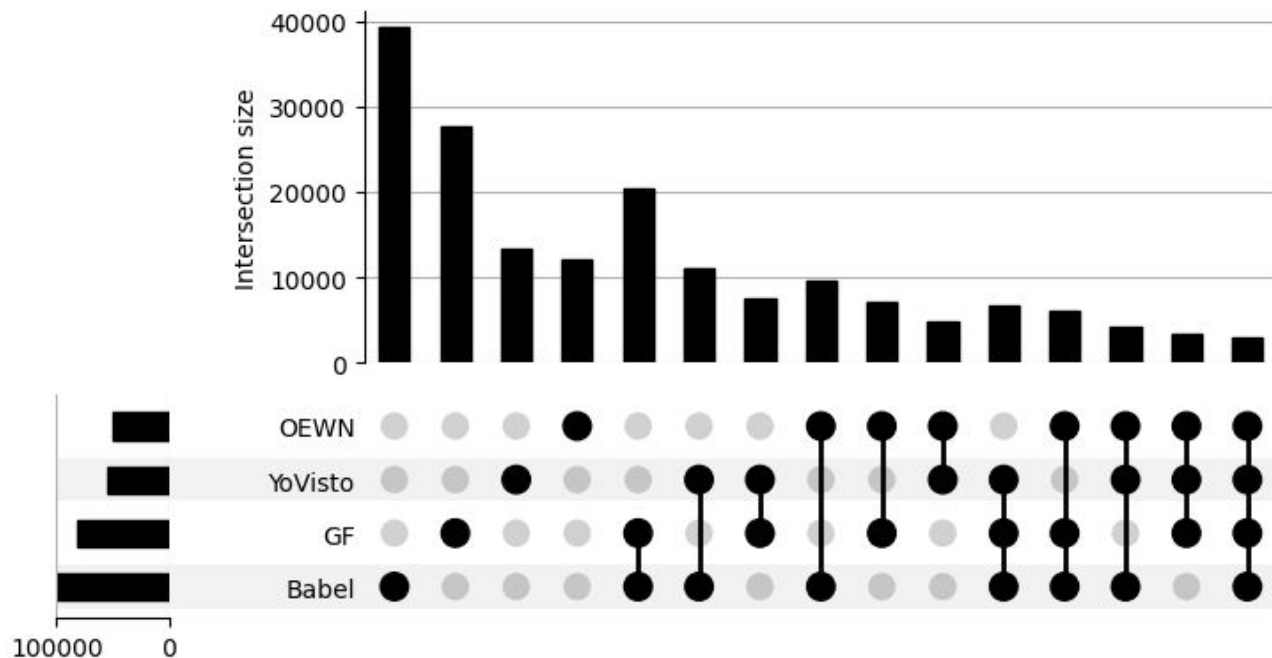
European rabbit
Temporal range: **Chibanian–Recent**^[2]
~0.6–0 *Ma*

PreЄ	Є	Є	S	D	C	W	P	J	K	PgN
------	---	---	---	---	---	---	---	---	---	-----

A photograph of a European rabbit (Lepus europaeus) sitting on a patch of green grass. The rabbit has brown and grey fur, large upright ears, and is looking towards the right of the frame. The background is a soft-focus green field.

Linking with Wikidata (Comparative size of the resources)

Joint work with John McCrae and Johann Bergh



Location and People Names from Wikidata

For NLG purposes the grammar is extended with names

WordNet	adjectives, nouns, verbs, etc.	100 thousand	
Wikidata	Given names	64 thousand	Describing 7.3 million people
	Family names	531 thousand	
	Place names	3.7 million	
	total	4.3 million	

Lookup for Abstract Expressions

Web-based shell at:

<https://cloud.grammaticalframework.org/wordnet/gf-functions.html>

- An entity in Wikidata is identified by QIDs, e.g. Q34 is Sweden, Q1 is the universe
- A simple API to the abstract expression (lexical item) for a QID:
 - `expr "Q34"`
 - `mkNP theSg_Det (expr "Q1")`

Constructions (Multiword Expressions)

abs: UseN (CompoundN square_1_N kilometre_1_N)

fre: kilomètre carré

spa: kilómetro de cuadrado

swe: kvadratkilometer

fin: neliökilometri

key: Q712226

abs: AdjCN (PositA square_1_A) (UseN kilometre_1_N)

key: Q712226

Retrieval of Constructions by QID

- `expr "Q712226" => square kilometer, kvadratkilometer`
- `expr "Q45068" => Carl Gustav`
- `expr "Q935" => Isaac Newton, Исаак Ньютон`
- `expr "P69" => \uni -> AdvVP (UseV graduate_V) (PrepNP from_Prep uni)`

Querying for Entities

```
let e = entity "Q34"  
in mkCl (mkNP (expr "Q34"))  
  (mkNP aSg_Det (mkCN (mkCN (expr e.P31.id))  
    (mkAdv with_Prep  
      [select: -1 | <mkNP (mkDecimal e.P1082.amount) inhabitantMasc_1_N,  
        e.P1082.P585.time>  
      ]  
    )))
```

Nondeterminism

- A property in Wikidata can have multiple values
- If a property value is missing then it corresponds to an empty variant

Control Structures

Make it possible to manipulate and aggregate variants:

[<keyword>: <opt. argument> | <expr. with variants>]

Examples:

- [select: -1 | <e.P1082.amount, e.P1082.P585.time>]
- [list: and_Conj | mkCN (expr e.P37.id)]

Content Planning

```
let e = entity "Q34"  
  cn0 = mkCN (expr e.P31.id)  
  cn1 = mkCN cn0  
    (mkAdv with_Prep  
      [select: -1 | <mkNP (mkDecimal e.P1082.amount) inhabitantMasc_1_N,  
                          e.P1082.P585.time>  
      ])  
in [one | (cn1 | cn0)]
```

Markup

First Class Support for Markup

```
let e = entity "Q34"
in <div>
  <h1>expr "Q34"</h1>
  <p>mkCl (mkNP (expr "Q34"))
    (mkNP aSg_Det (mkCN (mkCN (expr e.P31.id))
      (mkAdv with_Prep
        [select: -1 | <mkNP (mkDecimal e.P1082.amount)
          inhabitantMasc_1_N,
            e.P1082.P585.time>
        ]))
    ))</p>
</div>
```

Overloading, Polymorphism, Variants

QIDs correspond to different syntactic categories

- `expr "Q1" => universe_1_N : N`
- `expr "Q712226" => UseN (CompoundN square_1_N kilometre_1_N) : CN`
- `expr "Q935" => FullName isaac_1_GN newton_SN : NP`
- `expr "Q37" => sweden_LN : LN`

Polymorphism

- $\text{expr} : (\{A\} : \text{Type}) \rightarrow \text{Str} \rightarrow A$
- $\text{entity} : (\{A\} : \text{Type}) \rightarrow \text{Str} \rightarrow A$

Source Program:

```
let e = entity "Q34"  
in <e.P31.id, e.P1082.amount, e.P1082.P585.time>
```

Executable Program:

```
\{a\},\{b\},\{c\} ->  
let e = entity {{P31 : {id : a}; P1082 : {amount : b; P585 : {time : c}}}} "Q34" ;  
in <e.P31.id; e.P1082.amount; e.P1082.P585.time>
```

Overloading and Variants

mkNP (expr "Q34")

```
variants {(\np -> np) (expr {NP} "Q34");  
          (\n -> MassNP (UseN n)) (expr {N} "Q34"); MassNP (expr {CN} "Q34");  
          (\d -> UseDAP (DetDAP d)) (expr {Det} "Q34");  
          (\q -> UseDAP (DetDAP (DetQuant q NumSg))) (expr {Quant} "Q34");  
          UsePron (expr {Pron} "Q34");  
          UseLN (expr {LN} "Q34");  
          UsePN (expr {PN} "Q34")}
```

User Adaptations

Options

```
let qual = option quality_1_N of {  
    italian_A ;  
    swedish_A ;  
    delicious_1_A  
};
```

```
item = option food_1_N of {  
    apple_1_N ;  
    wine_1_N ;  
    pizza_N  
}
```

```
in mkNP (mkCN qual item)
```

Linguistic Variations

The RGL doesn't always produce the best result

```
case lang of {
  "fin" => mkPhrMark (mkCI (mkNP (mkQuant it_Pron) NumPl abutterMasc_N) neighbours);
  "rus" => mkPhrMark (mkCI (mkNP it_Pron) (mkVP (mkVP border_5_V) (mkAdv with_Prep neighbours)));
  "fre" => mkPhrMark (mkCI (mkNP theSg_Det country_2_N) (mkVP have_1_V2
    (mkNP num (mkCN (mkCN border_1_N) (mkAdv with_Prep neighbours)))));
  "bul" => mkPhrMark (mkCI (mkNP (ProDrop she_Pron)) (mkVP have_1_V2
    (mkNP num (mkCN (mkCN border_1_N) (mkAdv with_Prep neighbours)))));
  "spa" => mkPhrMark (mkCI (mkNP (ProDrop it_Pron)) (mkVP have_1_V2
    (mkNP num (mkCN (mkCN border_1_N) (mkAdv with_Prep neighbours)))));
  _      => mkPhrMark (mkCI (mkNP it_Pron) (mkVP have_1_V2
    (mkNP num (mkCN (mkCN border_1_N) (mkAdv with_Prep neighbours)))));
}
```

GF Pedia

GF Pedia 2.0

<https://cloud.grammaticalframework.org/wordnet/gf-wikidata.wiki>



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Sweden

Sweden is a Nordic country with 10,161,797 inhabitants. It has borders with Finland to the East, Norway to the West and Denmark to the South. The area is 447,425.16 square kilometres. A capital is Stockholm. Swedish, Finnish and Yiddish are official languages but Swedish, Finnish and Yiddish are also spoken.

demography

The life expectancy is 79.44 years. The fertility is 1.57 children per woman. The suicide rate stands at 12.4 deaths per 100,000 inhabitants yearly. Stockholm is the largest city in Sweden with 978,770 inhabitants. Sweden establishes the age of majority at 18 years. The minimum age of marriage is 18 years.

education

Education is obligatory for children from 6 years to 16 years. This results in a literacy rate of 99.9%. 9,424 children are out of the education system. This amounts to 0.09% of the population.

administrative units

the country has the following administrative units :

• Blekinge County • Dalarna County • Gotland County • Gävleborg County • Halland County • Jämtland County	• Jönköping County • Kalmar County • Kronoberg County • Norrbotten County • Skåne County • Stockholm County	• Södermanland County • Uppsala County • Värmland County • Västerbotten County • Västernorrland County • Västmanland County	• Västra Götaland County • Örebro County • Östergötland County
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politics

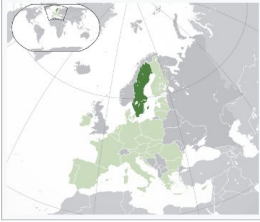
Sweden is a monarchy with Carl-Gustav as head of state. He succeeded Oscar in the position. The current head of government is prime minister Ulf Kristersson, which took office after



flag



coat of arms



TODO List

- A list of small bugs to fix
- Stabilize the user adaptation API
- Allow recursive function calls with Agda style termination checking
- Make it possible to mix languages in a single article
- Implement example based grammar writing
- ...

So far so good!